the others are unstable and emit either electrons or positrons (or decay by *K*-capture).

2. When the mass number as well as the number of neutrons and the number of protons are all even figures—in other words, in the case of ‘doubly even’ nuclei—usually there are several but not many (say, two or three) stable nuclei with the same mass number.

3. Stable nuclei which have an even mass number but an odd number of protons and an odd number of neutrons—the ‘doubly odd’ nuclei—are likely to be non-existent as a rule.

Exceptions to the third rule mentioned, however, occur among the very lightest nuclei. These can be accounted for by a particularly sharp flexure of the two curves, the result of which is that the most stable nucleus, situated approximately at the lowest point of the upper curve, lies below the nearest nuclei of the lower curve.

Otherwise, all the above conclusions are borne out almost completely by actual experience, as will be evident by reference to Table IV (at the end of the book), in which stable nuclei are indicated by black dots, the unstable ones by upright triangles (electron emitters) or inverted triangles (positron emitters). A careful study of this table will show that, in fact, except for a very few instances, there is never more than one stable nucleus for any given mass number. This statement is known as *Mattauch’s rule.* In this table, too, atoms having the same mass number are situated on a straight line, ascending to the left at an angle of 45 deg. to the abscissa. Thus, for instance, the palladium nucleus ${}_{46}Pd^{111}$, the silver nucleus ${}_{47}Ag^{111}$, the cadmium nucleus ${}_{48}Cd^{111}$ and the indium nucleus ${}_{49}In^{111}$ are all situated on such a straight line. Of these, only the ${}_{48}Cd^{111}$ is a stable one; the ${}_{46}Pd^{111}$ and ${}_{47}Ag^{111}$ nuclei are electron emitters, while the ${}_{49}In^{111}$ nucleus is a positron emitter. Thus this table fully confirms the fact that on the same straight line with a stable nucleus of an odd mass number all other nuclei of the same mass number are unstable. There is an exception to this rule, in the case of the mass number 113. In addition to the stable indium nucleus ${}_{49}In^{113}$ there is also a stable cadmium nucleus, ${}_{48}Cd^{113}$. This exception can probably be accounted for by the fact that both these nuclei happen to be situated at an approximately equal height on both sides of the energy surface, and the difference in energy between them is too small to permit the formation of an electron and a neutrino, or to admit of any measurable probability of the capture of an electron, so that a change of one of these nuclei into the other cannot occur. There may be also other exceptions, but their existence has not so far been confirmed by experience.

When the mass number is even, the occurrence of several stable nuclei having the same mass number is the general rule. Such nuclei are called